DWDM Lab by Usman Shehzaib
Lab Task Week # 7.
<u>Cube Deployment with measures and Calculations</u>

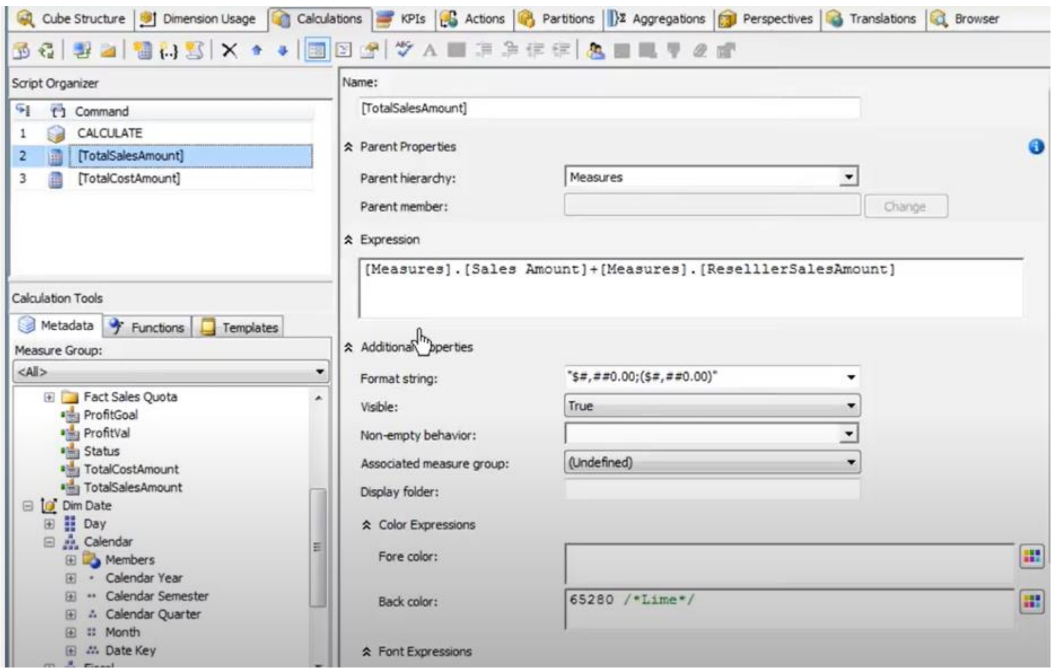
In this Week you will learn how to deploy Cube and add measures and calculations. Remember to start SSMS and SQL Server 2019 before you proceed.
Following tasks will be performed on a DSV which you have already created.

1. Click on New cube
2. Use existing tables and Factinternetsales, FactResellersales
3. Next from measures Select order quantity and sales amount and select same attributes from reseller sales table
4. Select all three dimensions
5. On next screen uncheck all dimensions
6. Save cube with a name and finish
7. Two fact tables are shown in yellow
8. Rename measures to Resellerorderquantity, Reseller salesamount

9. Process it and it will be shown on SQL analysis server instance and can be checked in Management studio
10. Go to Browser drag and drop sales amount in grid area also you can check by using different dimensions
11. Go to Dimension Usage Tab and check dimensions and measures where it shows how attributes are linked. Grey Box show there is no link
12. Also you can format measures go to properties of order quantity and in Format string put #,# it will display value as a thousand separator. Also for sales amount make it as $#,# it will make it currencyamount.
13. Now Lets add a calculated member.
14. First add a new measure there are different aggregations possible
15. Lets keep it Sum and name it as TotalproductCost and rename it Cost amount
16. Add another measure as totalproductcost and rename it reseller cost amount
17. Add a new measure group and add Fact sales quota. Here you find 4 different measures just keep sales amountquota
18. Now we should have 3 different measure groups
19. Go to fact salesquota properties and make ignoreunrelateddimenson to False this setting is done for null handling.
20. Process cube and browse it from SQL server. Here evaluate different measures

How to Add Calculations to your Cube

21. Now add calculations. For this go to calculation tab make a new calculation and name it TotalSalesAmount



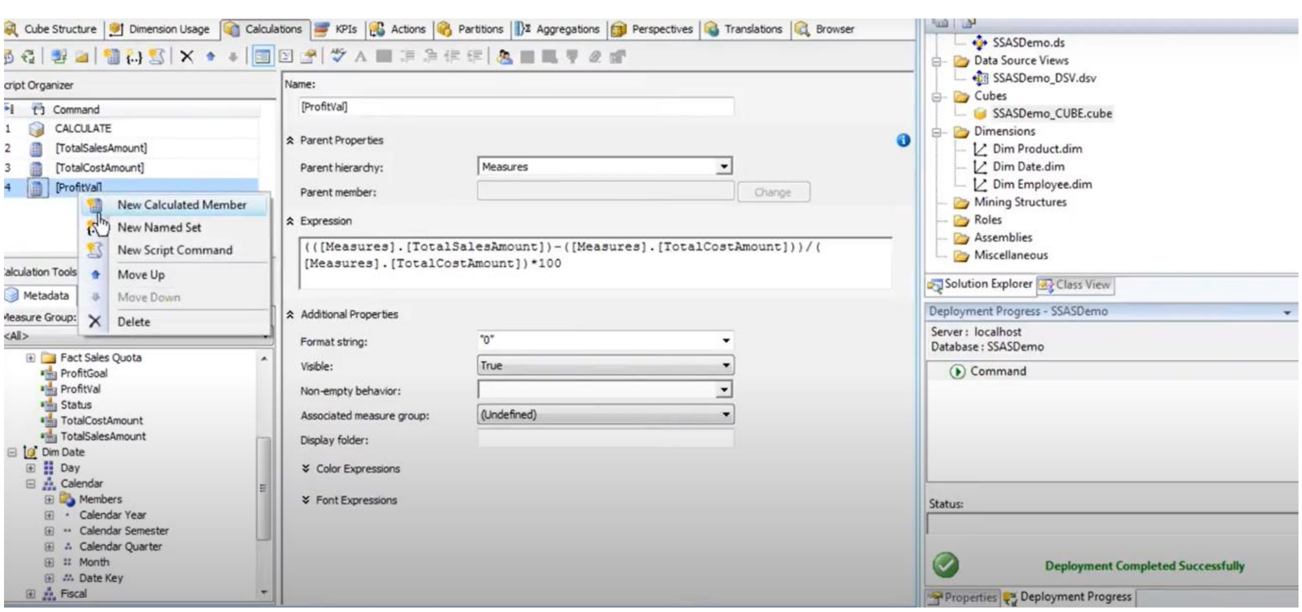
22. In Expression field put

[Measures] . [sales Amount] + [Measures] . [ResellerSalesAmount]

23. Set formattings as per choice

24. Same way we can add another measure like TotalCostAmount by summing total cost of both fact tables.

25. Now add ProfitVal Calculation using formula shown in figure



26. Now add another calculation named Goal and give it a value of 10 in expression field

27. Now add Status and set its expression as following three cases

```
Status

CASE WHEN [Measures].[Profitval] <0 THEN -1
WHEN [Measures].[Profitval] >= 0 AND [Measures].[Profitval] < 10
THEN 0
ELSE 1
END
```

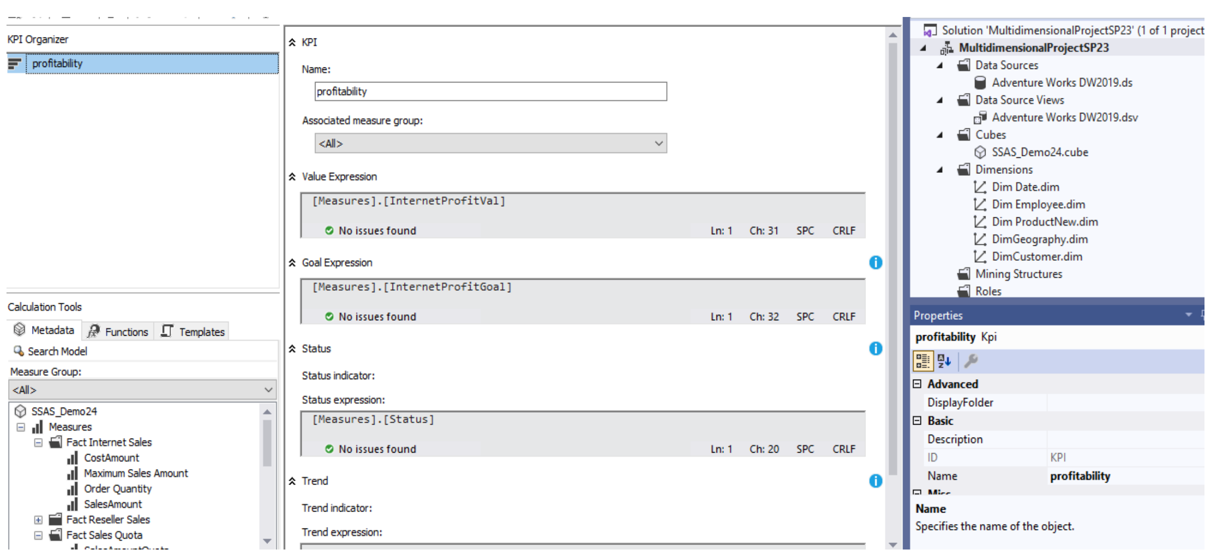
28. Make a trend calculation by using following formula

```
Trend
CASE WHEN ([Measures].[Profitval])<([Dim Date].[Calendar].PREVMEMBER,[Measures].[Profitval]) THEN -1
WHEN ([Measures].[Profitval])=([Dim Date].[Calendar].PREVMEMBER,[Measures].[Profitval]) THEN 0
ELSE 1
END
```

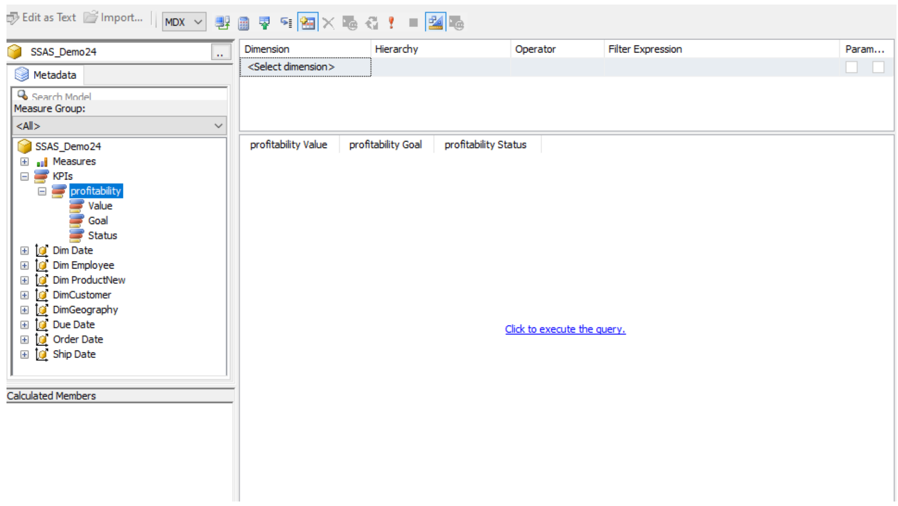
29.Now Process the Cube and check different calculations by browsing through different dimensions

Adding KPIs

1. Now lets add a KPI named Profitability in following way



2. Now Check KPIs in following way


Edit as Text
Import...
MDX
SSAS_Demo24
Metadata
Search Model
Measure Group:
<All>
SSAS_Demo24
Measures
KPIs
profitability
Value
Goal
Status
Dim Date
Dim Employee
Dim ProductNew
DimCustomer
DimGeography
Due Date
Order Date
Ship Date
Calculated Members
Dimension
Hierarchy
Operator
Filter Expression
Param...
<Select dimension>
profitability Value
profitability Goal
profitability Status
Click to execute the query.